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Effects of Nonuniform Inlet Velocity Profiles on Flow Regimes and Performance in Two-Dimensional Diffusers[1]

An analytical and experimental study on the effects of large distortions of inlet velocity profiles on flow regimes and performance in diffusers is reported. Experiments are restricted to flow in straight, two-dimensional diffusers with turbulent boundary layers. Systematic data are obtained for two general types of inlet flows: (1) simple, uniform shear flows in the core, and (2) severely nonuniform shear flows of the wake, jet, and step-shear type.

For uniform shear flows a first order prediction method based on inviscid rotational flow and the boundary layer blockage concept is developed and verified for diffusers operating in the unstalled flow regime. For nonuniform shear flows the inviscid rotational model is shown to predict performance trends better than the irrotational model; however, the inviscid rotational model is inadequate as a precise prediction method because no account is taken of mixing in the core region. Geometry and performance correlations for peak pressure recovery (at constant N/W_1) are also established.

Introduction

A classical diffuser problem is to determine the static-pressure rise for a diffuser of prescribed geometry having an arbitrarily specified inlet flow. It is well known that all possible variations of inlet conditions are not of equal importance in determining diffuser performance and flow. For example, unstalled and slightly stalled diffusers are rather weakly affected by variations in Mach number and Reynolds number based on mass flow average inlet conditions provided that: (*a*) inlet Mach number is in the low subsonic range and (*b*) inlet Reynolds number is greater than about 5×10^4. For the data and discussions to be presented here, these latter inlet flow conditions are met or assumed. For these cases the most important inlet condition that affects performance is the shape of the inlet velocity profile.[1]

Many researchers have worked on diffuser problems and much has been accomplished for the restricted class of flows having symmetrical, uniform inlet velocity profiles over a core region

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[1] The present paper is taken primarily from the report of Wolf and Johnston [19] and the PhD thesis by Wolf of the same title, Stanford University, Dec. 1966.

Nomenclature

A = cross-sectional area

A_B = blocked area, $1 - A_E$

A_E = effective area, $\int_A \frac{u}{u_{n\,ax}} dA$

AR = diffuser total area ratio, A_2/A_1

B = total blockage factor, $A_B/A = 1 - E$

B_B = boundary layer blockage factor, $(\delta_L^* + \delta_R^*)/W$

b = height of two-dimensional diffuser between parallel walls

Cp = measured static-pressure recovery coefficient, $\frac{\bar{P}_2 - \bar{P}_1}{\bar{q}_1}$

$Cp(x)$ = measured local static-pressure recovery coefficient, $\frac{P(x) - \bar{P}_1}{\bar{q}_1}$

Cp_{1-D} = pressure recovery based on inviscid one-dimensional flow,

$$\frac{P_2 - P_1}{\rho \bar{u}_1^2/2} = 1 - \frac{1}{AR^2}$$

Cp_i = pressure recovery based on inviscid flow having an idealized inlet velocity profile designated as u_{i1}, $\frac{\bar{P}_2 - \bar{P}_1}{\rho u_{i1}^2/2}$

Cp_{ic} = pressure recovery based on inviscid flow with uniform shear having an idealized inlet velocity profile designated as u_{ic1}, $\frac{\bar{P}_2 - \bar{P}_1}{\rho \bar{u}_{ic1}^2/2}$

Cp_m = pressure recovery based on reversible mixing of the flow from a nonuniform inlet velocity profile to a uniform inlet velocity profile followed by inviscid one-dimensional diffusion to the exit section,

$$\frac{\bar{P}_2 - P_1}{\rho \bar{u}_1^2/2} = K_{E1} - \frac{1}{AR^2}$$

E = effective area fraction, A_E/A

K_E = kinetic-energy flux coefficient, $\frac{1}{A}\int_A \left(\frac{u}{\bar{u}}\right)^3 dA$

K_M = momentum flux coefficient, $\frac{1}{A}\int_A \left(\frac{u}{\bar{u}}\right)^2 dA$

N = axial length of diffuser

P = static pressure

$\bar{P}$ = average static pressure—for the present data P is the arithmetic average of the static-pressure measurements

q = dynamic pressure based on u_{1avg}, $\rho u^2_{1avg}/2$

U = nondimensional velocity u/u

u = velocity in x-direction

$\bar{u}$ = area-averaged velocity for three-dimensional flow, $\frac{1}{A}\int_A u dA$ for two-dimensional flow, $\int_0^1 u d\zeta$

v = velocity in y-direction, or velocity normal to x-axis

(Continued on next page)

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outside the wall boundary layer regions (see Fig. 1(*a*)). Major accomplishments include: (*a*) classification of flow regimes [1, 2][2]; (*b*) construction of performance maps [3]; (*c*) development of performance prediction methods based on boundary layer theory [4, 5], and (*d*) recent correlation by Sovran and Klomp of the effects of boundary layer type of inlet velocity non-uniformity on the basis of inlet displacement thickness effects generalized in the concept of area blockage [6].

The foregoing accomplishments have been of immense aid in diffuser design. However, in many applications the simple, irrotational core flows used for almost all of the reported work are not obtained because of various upstream flow conditions, such as, partial flow blockage caused by wakes shed from solid obstacles (struts, blades, rotors, etc.). Thus, there is considerable need to determine the effects of non-uniformities in the core region of the inlet velocity profile on diffuser flow and performance. Several researchers have worked on this problem [7, 8, 9, 10]. Their results show that inlet core-flow velocity-profile distortions cause significant changes in pressure-recovery performance, and by implication, flow regime. Some profiles such as isolated jets may cause reduced performance, but others, such as isolated wakes, may result in apparent increases in performance over that measured for uniform-core inlet velocity profiles in the same diffuser. However, the total experimental and theoretical information uncovered prior to this study was insufficient to serve as a basis for a practical design method or for the development of a prediction method for evaluating the effects of nonuniform inlet conditions on diffuser flow and performance.

The general objective of this work is to provide better understanding of the effects of nonuniform inlet velocity profiles on diffuser flow and performance. Specific objectives are:

1 To develop appropriate models for the ideal diffusion process;

2 To produce data covering a wide range of two-dimensional straight-walled diffuser geometries to serve as a basis for design; and

3 To lay the ground work for future development of performance prediction methods.

To achieve the above objectives, two general types of flow are considered:

1 Simple, uniform shear flows, i.e., flows with constant vorticity in the core (see Fig. 1(*b*)); and

2 Severely nonuniform shear flows including wakes, jets and step-shear flows (see Fig. 1(*c*)).

[2] Numbers in brackets designate References at end of paper.

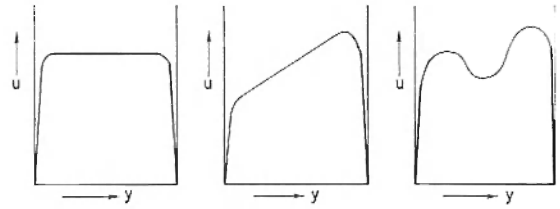

Fig. 1 Illustration of three types of inlet flows

Ideal Diffusion Process

Considerable progress has been made on problems involving free (unbounded) shear flow with nonuniform transverse velocity gradients, including effects of viscous shear stress. A general summary of work up to 1964 has been reported by Halleen [11] for the far field region of turbulent flow cases. Specific investigations on special types of nonuniform flow in axial pressure gradients have been reported by several investigators [12, 13, 14, 15, 16, 17]. None of these reports, however, provide the type of information required to gain understanding of diffuser performance with generalized inlet conditions.

As a first step in developing this understanding consider the ideal flow problem. An inviscid solution for diffuser pressure recovery can be obtained for the situation in which the flow is divisable into an inviscid core region and thin viscous boundary layer regions. This solution provides the boundary conditions that must be used in conjunction with boundary layer theory to solve the flow problem near the wall. In certain cases, the inviscid solution for the complete flow field (i.e., neglecting the wall boundary layers) may serve as a first order approximation to the actual behavior. For these reasons the ideal diffusion process to be described is based on steady, incompressible, inviscid, rotational flow, i.e., a flow for which the circulation set by the given inlet conditions is preserved along streamlines.

Two general methods for solution of the inviscid core flow problem are presented in reference [19]: 1, a differential-equation method limited to two-dimensional core flows entering two-dimensional diffusers and 2, an integral-equation method of more general nature that can be applied to cases with three-dimensional, nonuniform shear flows at the inlet of straight diffusers of arbitrary cross-sectional shape. Only the latter, more general method is presented here.

Nomenclature

W = width of two-dimensional diffuser between diverging walls

x = axial or flow direction measured from the diffuser throat

y, z = distances normal to x-direction to form a right-hand coordinate system

Δ = denotes difference

δ^* = boundary layer displacement thickness

ζ = nondimensional coordinate normal to the axial (x) direction, $(y + W/2)/W$

η_{1-D} = effectiveness based on one-dimensional, inviscid flow, Cp/Cp_{1-D}

η_i = effectiveness based on inviscid flow having an ideal velocity profile designated as u_{i1}, Cp/Cp_i

η_{ic} = effectiveness based on inviscid flow having an ideal velocity profile designated as u_{ic1}, Cp/Cp_{ic}

λ = velocity parameter defined for simple, uniform shear flows as u_{icmax}/u_{icmin} at diffuser inlet

ρ = fluid density

Subscripts

1 = diffuser inlet

2 = diffuser exit

avg = after velocity indicates average velocity, (mass flow rate)/ρA

c = refers to uniform shear flow

f = refers to forward flow limit predicted by inviscid flow model

i = after velocity refers to idealized velocity profile—defined as the actual velocity in the core-flow region and equal in value to the velocity at the outer edge of the boundary over the wall boundary layer region

ic = after velocity refers to idealized velocity profile—defined for simple, uniform shear flows as the actual core profile extended to the walls at the fixed vorticity of the core region

L = diverging wall on left-hand side of diffuser (for simple, uniform shear flows this designates the low velocity side)

max = refers to maximum value

min = after velocity refers to minimum value of velocity profile over the core region

R = diverging wall on right-hand side of diffuser (for simple, uniform shear flows this designates the high velocity side)

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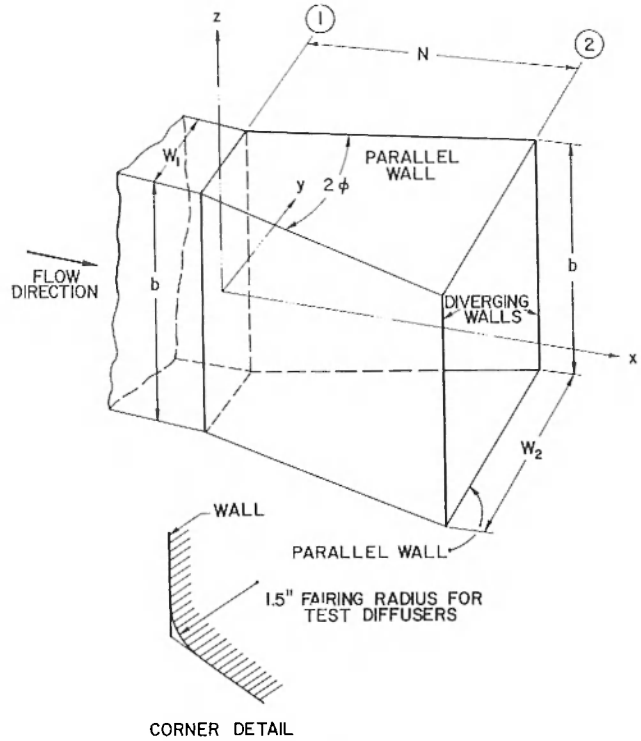

Fig. 2 Two-dimensional diffuser geometry with nomenclature

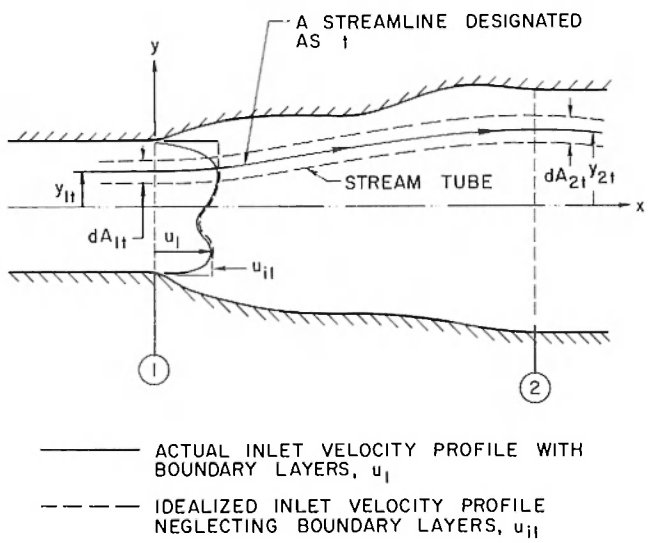

Fig. 3 Longitudinal cross section of general diffuser system of arbitrary transverse cross section

Fig. 2 shows the geometry and nomenclature of a typical two-dimensional diffuser used in the testing program to be described in a later section. The diffuser considered in the analysis is of more general shape as illustrated in Fig. 3. The cross-sectional shape of the diffuser in Fig. 3 can be arbitrary as long as contours are relatively smooth, especially near inlet and outlet where the walls are assumed to be nearly parallel. An idealized inlet velocity profile, u_{i1}, see Fig. 3, is assumed so that the velocity does not go to zero at the walls. This idealization is necessary since an ideal analysis based on a real inlet velocity profile, u_1, with zero velocity at the walls, would result in a backflow region near the walls immediately downstream of section 1 to allow for a static-pressure rise. A typical streamline, t, within a stream tube is shown in Fig. 3. The ideal flow assumptions allow the use of the Bernoulli equation to relate static pressure and velocity on the t-streamline between inlet, section 1, and outlet, section 2, as

$$P_{1t} + \frac{\rho}{2}(u_{i1t}^2 + v_{i1t}^2) = P_{2t} + \frac{\rho}{2}(u_{i2t}^2 + v_{i2t}^2) \qquad (1)$$

where u_{it} designates the main, or axial, velocity component and v_{it}, the component of velocity normal to the x-axis. For most high performance, straight diffusers the angle of divergence between the side walls is small and the walls are relatively straight. Thus, to a good approximation, the velocity vectors at diffuser inlet and outlet are nearly parallel[3] and so the normal velocity components, v_{it}, are small compared to the axial components, u_{it}. Also, as the streamline curvature is negligible, the static pressure will be nearly constant from streamline to streamline at a given station, i.e., $P_{1t} = P_1$ and $P_{2t} = P_2$. With these approximations equation (1) becomes

$$P_2 - P_1 = \frac{\rho}{2}(u_{i1t}^2 - u_{i2t}^2) \qquad (2)$$

An ideal static-pressure recovery coefficient of the diffuser, Cp_i, can now be expressed in terms of velocities along a given streamline by utilizing equation (2):

$$Cp_i \equiv \frac{P_2 - P_1}{\frac{\rho}{2}\bar{u}_{i1}^2} = \left(\frac{u_{i1t}}{\bar{u}_{i1}}\right)^2 - \left(\frac{u_{i2t}}{\bar{u}_{i1}}\right)^2 = U_{i1t}^2 - \left(\frac{U_{i2t}}{U_{i1}}\right)^2 \qquad (3)$$

Here, $\bar{u}_{i1}$, is the area-averaged ideal inlet velocity and U_{i1} is defined as a dimensionless velocity,

$$U_{i1} = \frac{u_{i1}}{\bar{u}_{i1}} \qquad (4)$$

To obtain an expression for Cp_i in terms of the inlet velocity profile and the diffuser area ratio, $AR = A_2/A_1$, the continuity equation is first applied to the stream tube surrounding the t-streamline, i.e.,

$$u_{i1t}dA_{1t} = u_{i2t}dA_{2t} \qquad (5)$$

The desired relationship for Cp_i can now be found in terms of an integral equation by combining equations (3) and (5) and integrating over all stream tubes entering the diffuser; for the assumption of no backflow at either section 1 or 2, one obtains

$$AR = \frac{1}{A_1}\int_{A_1} \frac{U_{i1}dA_1}{\sqrt{U_{i1}^2 - Cp_i}} \qquad (6)$$

The t-subscript has been dropped since the integration is performed over the entire inlet area where U_{i1} is specified for each element of area dA_1.

If the diffuser inlet area, inlet velocity distribution and area ratio are given, then equation (6) is a nonlinear, singular integral equation for the unknown pressure-recovery coefficient, Cp_i. The equation has a real solution provided that the denominator of the integrand is positive everywhere at the inlet plane, i.e.,

$$(U_{i1}^2 - Cp_i) > 0 \qquad (7)$$

The condition expressed in equation (7) imposes a maximum allowable value on Cp_i for a given inlet velocity profile and area ratio, AR. The significance of this limiting condition is explained as follows. At least one streamline will have minimum inlet velocity, $U_{i1\min}$. As the pressure rises in the diffuser, it is this streamline that will first exhibit zero velocity. The axial, x, location where the streamline of minimum inlet velocity first goes to zero is denoted as the *forward flow limit*.[4] The ideal recovery, Cp_{if}, at the forward flow limit is the maximum ideal recovery attainable. This fact can be deduced from equation (3) when the value of u_{i2t} is set equal to zero.

$$(Cp_i)_{\max} = Cp_{if} = U_{i1\min}^2 \qquad (8)$$

[3] The absence of swirl flow about the x-axis is assumed. Significant swirl would negate the results of this analysis.

[4] The term forward flow limit is used to distinguish this theoretical limit from the inception of separation in real flow.

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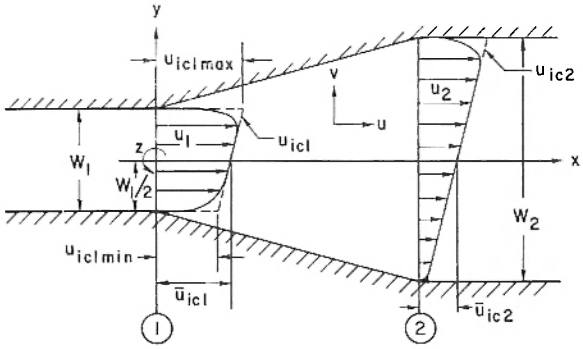

Fig. 4 Uniform shear flow in two-dimensional, unstalled diffuser

With respect to equation (6) this limit implies that real solutions exist only for $Cp_i < Cp_{if}$. The value of the area ratio at the forward flow limit, AR_f is obtained by substituting Cp_{if} for Cp_i in equation (6).

$$AR_f = \frac{1}{A_1}\int_{A_1} \frac{U_{i1}dA_1}{\sqrt{U_{i1}^2 - U_{i1min}^2}} \tag{9}$$

Real diffusers can have area ratios in excess of AR_f. In these cases Cp_i is assumed arbitrarily to equal the maximum value of Cp_i, i.e., $Cp_i = Cp_{if}$ for all AR greater than AR_f.

Since the theory of nonlinear integral equations is very limited, no general inversion technique for equation (6) was discovered. However, a number of interesting special cases are solved in closed form in reference [19]. The equation can, for practical purposes, be readily solved by numerical integration. Once the ideal inlet profile, U_{i1}, is specified, the calculation is performed by evaluating the area ratio, AR, for preassigned values of Cp_i selected between 0 and Cp_{if}.

When the diffuser geometry and inlet flow, Fig. 4, are two-dimensional, equation (6) can be recast in terms of dimensionless variables as follows

$$AR = \int_0^1 \frac{U_{i1}(\zeta_1)d\zeta_1}{\sqrt{U_{i1}^2(\zeta_1) - Cp_i}} \tag{10}$$

where ζ_1 is the dimensionless distance from the diffuser center line,

$$\zeta_1 = \frac{y_1 + W_1/2}{W_1}$$

A simple and important special case of two-dimensional flow that can be solved in closed form is that of *uniform shear flow* (see Figs. 1(*b*) and 4). The constant core-flow slope in this case is used to define the ideal inlet and outlet velocity profiles u_{ic1} and u_{ic2}, by linear extension of the core-flow profile to the walls. The additional subscript c is appended to all quantities when this special idealization of the inlet profile is made. If the subscript c is omitted, the ideal velocity in the boundary layer region is assumed constant and equal to the local core velocity adjacent to the boundary layer edge as shown in Fig. 3.

For the idealized uniform shear flow case it is easily shown that the slopes of the inlet and outlet profiles are identical because the vorticity is the same for all streamlines and constant along each streamline.[5] From this fact and by application of the conservation-of-mass principle to a control volume between the walls of the diffuser and extending from station 1 to 2, the value of outlet velocity on either diffuser wall is obtained. The ideal pressure-recovery coefficient, Cp_{ic}, is then determined from equation (2) as

$$Cp_{ic} \equiv \frac{P_2 - P_1}{\frac{\rho}{2}\bar{u}_{ic1}^2} = 1 - \frac{1}{AR^2} - \left(\frac{\lambda - 1}{\lambda + 1}\right)^2 (AR^2 - 1) \tag{11}$$

where λ is an inlet velocity profile parameter defined as the ratio of maximum to minimum inlet velocity, i.e., $\lambda = (u_{max}/u_{min})_{ic1}$. This expression for Cp_i can also be derived from equation (10).

The maximum ideal recovery attained at the forward flow limit and the corresponding area ratio are, respectively:

$$Cp_{icf} = \left(\frac{(u_{ic1})_{min}}{\bar{u}_{ic1}}\right)^2 = \left(\frac{2}{\lambda + 1}\right)^2 \tag{12}$$

$$AR_{fc} = \sqrt{\frac{\lambda + 1}{\lambda - 1}} \tag{13}$$

For all values of λ greater than unity the value of AR_{fc} is finite. If the ideal inlet velocity is assumed constant through the wall boundary layer regions, the value of AR_f is infinite. In fact, it can be shown in general that if the velocity gradient, $\partial u_i/\partial y$ on the streamline of minimum velocity is zero than AR_f is infinite; if, on the other hand, it is nonzero, as assumed for the uniform shear flow case, the area ratio at the forward flow limit will be finite.

For uniform shear flow no change of velocity profile slope will occur in the core even in the flow of real fluids. This is because there are no transverse gradients of shear stress to cause lateral transport of axial momentum in the core, i.e., there is said to be no "mixing" in the core.[6] In the real case the wall boundary layers will of course grow in thickness and reduce the lateral size of the core region. The velocity profiles at outlet in the real case will, consequently, always be more distorted than predicted by the ideal theory. Therefore, as increased distortion reduces pressure recovery, see Sovran and Klomp [6], the ideal recovery coefficient, Cp_{ic}, should provide a valid upper limit to the actual recovery coefficient, Cp, and the diffuser effectiveness defined as

$$\eta_{ic} = \frac{Cp}{Cp_{ic}} \tag{14}$$

should always be less than unity. Since the core vorticity is constant in real flows, η_{ic} can be assumed to be independent of core distortion (values of λ or core vorticity) and, to a first approximation for unstalled diffusers, equal to the η-values determined for uniform inlet velocity profiles in geometrically identical diffusers having the same inlet boundary layer blockage. This correspondence cannot be expected to hold after the onset of stall because of the difference between the ideal flow model which assumes $Cp_{ic} = Cp_{icf}$ for $AR > AR_{fc}$ and the one-dimensional model conventionally used in the definition of n for irrotational, inlet flows.

Experimental Results and Discussion

The experimental program was designed to obtain systematic data on flow regimes and performance in straight two-dimensional diffusers with nonuniform inlet velocity profiles and, in addition, to verify the validity of performance trends and limiting conditions found by the idealized diffusion analysis presented in the previous section.

Description of Tests. Diffuser performance was determined with the air apparatus previously described by Sagi, et al. [18]. The apparatus consists of a blower, appropriate ducting, damping screens for velocity-profile control, and a nozzle to accelerate the flow from a plenum chamber to the diffuser inlet section. The diffuser inlet dimensions are $b = 12$ in. and $W_1 = 3$ in. (see Fig. 5). Several modifications downstream of the nozzle section were necessary in order to incorporate velocity-profile generators for

[5] In two-dimensional, inviscid, incompressible flow vorticity is conserved along streamlines as vortex filaments that are perpendicular to the plane of flow do not change length as they are convected with the fluid.

[6] With nonuniform core distortions at inlet (wakes, jets, etc.) there will be "mixing." That is, laminar or turbulent viscosity will tend to flatten core flow velocity profiles downstream of the inlet.

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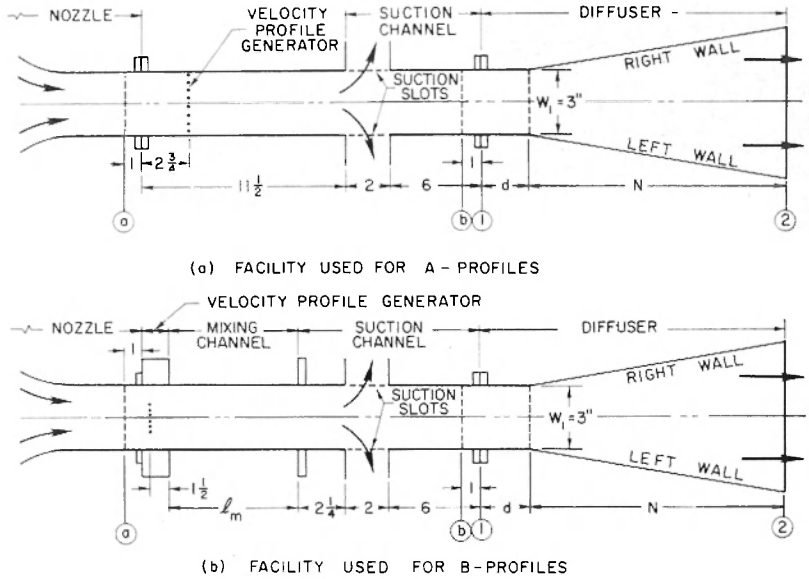

Fig. 5 Schematic representation of test section of diffuser test facility
(a) Facility used for A-profiles
(b) Facility used for B-profiles

providing specified core-flow distortions and a suction channel for controlling the diffuser inlet boundary layer thickness. The new parts were first designed as temporary equipment to study functional suitability and, subsequently, used for the first phase of the testing program with the uniform shear flows. Fig. 5(*a*) is a schematic representation of the temporary equipment. The final facility, designed to be more flexible than the original equipment, is shown in Fig. 5(*b*). For testing diffusers with the non-uniform shear flows, the variable length part of the channel, designated as l_m, was set equal to zero. All diffusers were built with a short throat length, designated as d, equal to about two inches. The velocity-profile generators were built by appropriately spacing fine wires held in tension in a metal framework

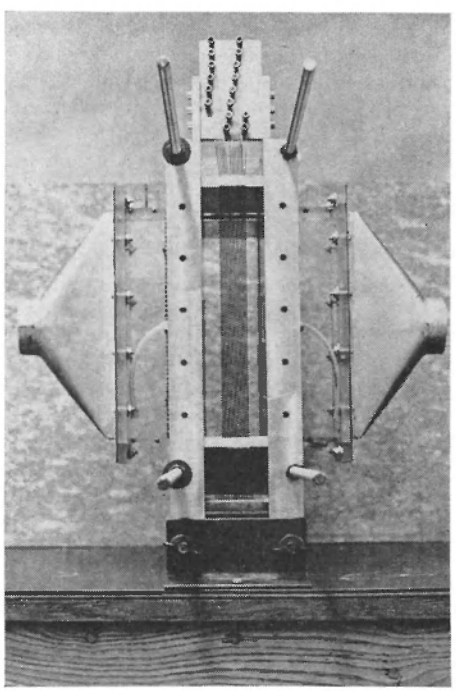

Fig. 6 Velocity-profile generator, Grid B-1, installed upstream of suction channel

[19]. Fig. 6 shows a typical grid installed in the channel section that fits between the nozzle of the test facility and the test diffuser.

Systematic data collected during the test program included observations of flow patterns and measurements of static-pressure distributions and velocity profiles. The diffuser flow patterns were determined by observing wool tufts fastened to the diffuser walls. The overall static-pressure recovery was determined according to the following equation:

$$Cp = \frac{\bar{P}_2 - \bar{P}}{\bar{q}_1}$$

where P_2 refers to the atmospheric pressure at the diffuser exit, section 2 (see Fig. 5), $\bar{P}_1$ refers to the diffuser inlet pressure as determined from 16 static-pressure taps located at section b, and $\bar{q}_1$ is the inlet dynamic pressure defined as $\rho u_{1\text{avg}}^2/2$. The experimental uncertainty in the pressure recovery is approximately as follows:

(*a*) For Cp between 0.4 and 0.7: ± 3 to ± 5 percent
(*b*) For Cp between 0.2 and 0.4: ± 5 to ± 7 percent.

Inlet velocity profiles were measured for each velocity-profile generator at section 1 and exit velocity profiles were measured at section 2. All velocity profile data reported in this paper were measured by traversing with a total-head probe along the transverse centerline and recording measurements every 0.1 inch in the core and every 0.01 in. in the thin boundary layer regions. Other profiles were taken to check on two-dimensionality of the flow which was excellent at the inlet section (± 1.5 percent variation outside the thin wall boundary layers), but rather poor at outlet especially for the stalled diffusers.

Diffuser Tests With Simple Uniform Shear Flows. Two sets of diffusers having N/W_1-values of 3 and 6 were selected for testing. The geometries, marked as x in Fig. 7, were selected to cover the flow regime chart near the line of first stall, line a-a.

Five uniform shear velocity profiles were generated with one wire grid by varying boundary-layer-suction flow rates. These profiles are designated as A-profiles for identification purposes. General characteristics of the profiles are listed in Table 1. The velocity profiles had a variation in the profile parameter λ of 1.50 to 1.73 and boundary layer blockage B_{Bc1} of 0.017 to 0.053. Profile A-1 was generated with no boundary layer suction; profile A-2, with suction on the low velocity wall; profile A-3 and A-4,

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Table 1 Characteristics of inlet velocity profiles

Velocity Profile Code Description	Re_{W_1}	λ	Vorticity (sec^{-1})	u_{max}/u_{cin}	u_{max}/u_{ave}	$\bar{u}^*/u_{ave}$	δ_M	K_E	B	Cp_{1f}	Cp_{icf}	AR_f	B_B	B_{Bc}
A-1 Uniform shear flow	2.0×10^5	1.50	229	1.28	1.176	0.990	1.028	1.073	0.155	0.763	0.640	2.24	0.054	0.053
A-2 Uniform shear flow	1.8×10^5	1.56	228	1.39	1.190	0.999	1.023	1.061	0.157	0.676	0.610	2.14	0.035	0.032
A-3 Uniform shear flow	1.8×10^5	1.73	235	1.58	1.280	1.007	1.032	1.087	0.210	0.615	0.537	1.93	0.017	0.017
A-4 Uniform shear flow	1.8×10^5	1.69	249	1.56	1.266	1.010	1.031	1.085	0.199	0.622	0.553	1.98	0.018	0.017
A-5 Uniform shear flow	1.8×10^5	1.70	257	1.49	1.232	0.997	1.029	1.080	0.188	0.644	0.549	1.97	0.030	0.041**
B-1 Wake	1.6×10^5	--	-	1.75	1.306	0.997	1.054	1.157	0.233	0.540	--	-	0.011	--
B-2 Jet	1.6×10^5	--	-	1.68	1.253	1.000	1.052	1.151	0.202	0.552	--	-	0.012	--
B-3 Step-Shear	1.6×10^5	--	-	1.73	1.292	1.001	1.065	1.193	0.225	0.546	--	-	0.012	--
B-4 Transverse wakes	1.6×10^5	--	-	--	--	--	--	--	--	--	--	-	--	--
- Uniform	1.6×10^5	1.00	0	1.07	0.96	1.000	1.005	1.012	0.022	1.000	1.00	-	--	--

*The values $\bar{u}$ and u_{ave} differ slightly as end wall displacement thickness is not included in $\bar{u}$. The ratio should exceed one by a small amount. Experimental uncertainty causes deviation.

**Refers to a blockage based on a mean boundary layer displacement thickness weighted over 75 percent of the wall height, b.

with suction on both diverging walls; and profile A-5, with suction on the end walls only. Fig. 8 compares the shape of the A-profiles; profile A-4 has been omitted because it is essentially identical with profile A-3.

Exit velocity profiles were measured for selected unstalled diffusers. A typical comparison of inlet and exit velocity profiles is presented for A-1 flow in Fig. 9. The exit core slope predicted on the basis of fixed vorticity is very close to the actual core slope. This agreement justifies the use of the inviscid rotational flow model for unstalled diffusers with nonmerging boundary layers and constant vorticity core flow at the inlet section.

From the tuft flow visualization studies it was observed that a stable two-dimensional stall developed on the low velocity side. The onset of stall occurs at a lower value of AR for fixed N/W_1 than for diffusers with irrotational inlet core velocity profiles. Line a-a and line b-b established by Kline and Fox [1] are shifted appreciably downward.

The pressure-recovery data for all diffusers tested with uniform shear flows are presented in Table 2. In Fig. 10 data collected for diffusers with A-1 inlet velocity profiles are compared to performance predicted by the ideal diffusion model, Cp_{ic}. Curves for the inviscid one-dimensional flow model, Cp_{1-D}, defined in the Nomenclature, and the performance predicted from the data charts of Reneau, et al. [3] for uniform core flow and equivalent inlet boundary layer blockage, B_{Bc1} are included for comparison. The performance trends are typical of those obtained for all the A-profiles. The differences between the Cp_{1-D} curve and the Reneau data are due to flow losses that occur in the real diffuser flow. Reneau, et al. have shown that for unstalled diffusers with

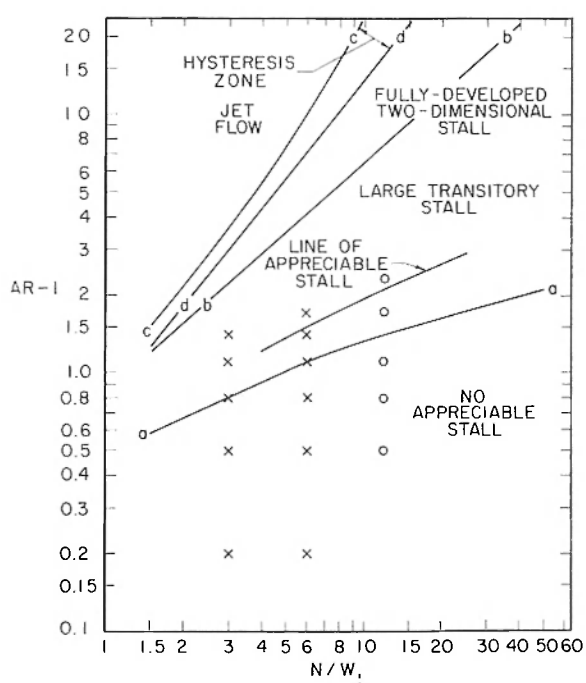

Fig. 7 Diffuser flow regime chart as established by Fox and Kline [1] for two-dimensional diffusers with uniform inlet velocity profiles, showing the geometries of the diffusers tested

Table 2 Performance of diffusers with uniform shear flows

Inlet Velocity Profile	Area Ratio AR	Measured Pressure Recovery Cp N/W₁ = 3	N/W₁ = 6
A-1	1.2	0.240	[illegible]
A-1	1.5	0.369	[illegible]
A-1	1.8	0.342	[illegible]
A-1	2.1	0.295	[illegible]
A-1	2.4	0.264	[illegible]
A-1	2.7	---	[illegible]
A-2	1.2	0.255	[illegible]
A-2	1.5	0.402	[illegible]
A-2	1.8	0.358	[illegible]
A-2	2.1	0.324	[illegible]
A-2	2.4	0.286	[illegible]
A-2	2.7	---	[illegible]
A-3	1.2	0.230	[illegible]
A-3	1.5	0.361	[illegible]
A-3	1.8	0.322	[illegible]
A-3	2.1	0.290	[illegible]
A-3	2.4	0.260	[illegible]
A-3	2.7	---	[illegible]
A-4	1.5	---	[illegible]
A-4	1.8	---	[illegible]
A-4	2.1	---	[illegible]
A-4	2.4	---	[illegible]
A-4	2.7	---	[illegible]
A-5	1.2	0.240	[illegible]
A-5	1.5	0.356	[illegible]
A-5	1.8	0.327	[illegible]
A-5	2.1	0.291	[illegible]
A-5	2.4	0.269	[illegible]
A-5	2.7	---	[illegible]

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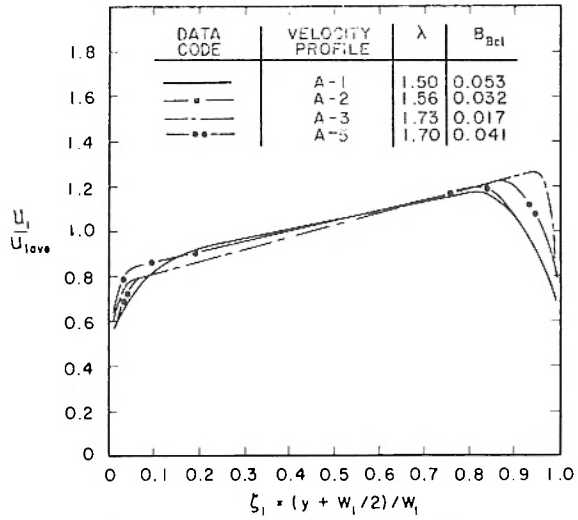

Fig. 8 Measured inlet velocity profiles of simple, uniform shear flows at constant vorticity; A-1, A-2, A-3, and A-5. Note: Curves represent best fit through data measured every 0.1 in. in the core and every 0.01 in. in the boundary layer.

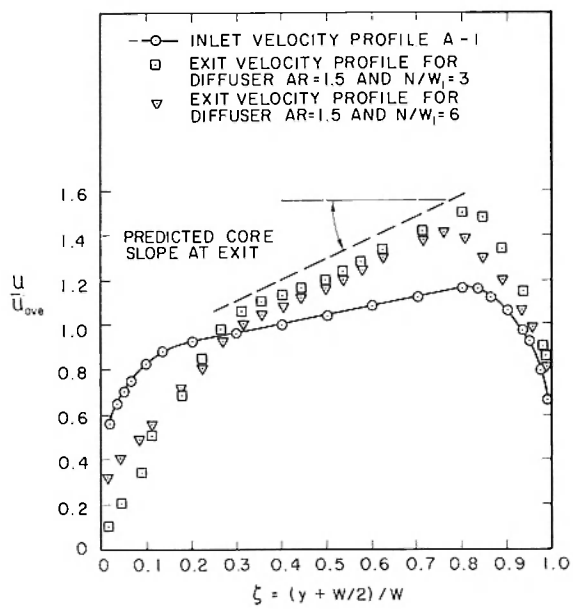

Fig. 9 Comparison of velocity profiles of a simple, uniform shear flow, A-1, at inlet and exit sections of two-dimensional diffusers

thin inlet boundary layers the effect of these losses on pressure recovery can be satisfactorily calculated from boundary-layer-growth theory. It is reasoned that the differences between the Cp_{ic}-curve and the authors' data are similar to those observed between the Cp_{1-D}-curve and the data of Reneau, et al. Consequently, it is expected that a prediction method for diffusers with *unstalled* uniform shear flows having thin inlet boundary layers could be based upon the present ideal diffusion model, Cp_{ic}, for the core region and an appropriate boundary layer growth prediction method for the wall boundary layer region.

In Fig. 11(*a*) and (*b*), the measured effectiveness values, η_{ic}, defined in equation (14), are compared to the η-values determined from the Reneau charts for uniform flow and equivalent inlet boundary layer blockage, B_{Bcl}. For unstalled flow, $AR \leq 1.5$, the Reneau curves tend toward the experimental curves. This indicates that for the unstalled flow cases the ideal diffusion

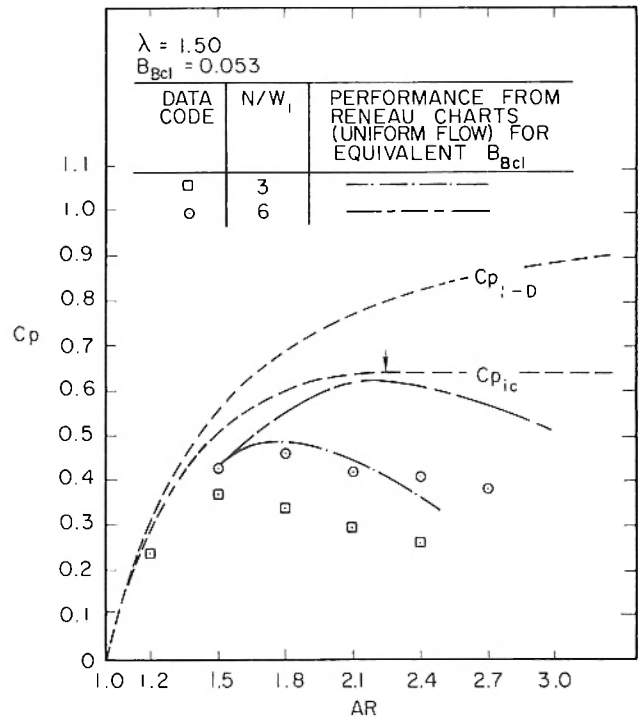

Fig. 10 Comparison of measured static-pressure recovery of two-dimensional diffusers with simple, uniform shear flow, A-1, and performance predicted by simple flow models. The arrow marks the forward flow limit condition determined by equation (13).

model, Cp_{ic}, is effectively taking account of core distortion and that the spread of the η_{ic}-curves is due to variation in the inlet boundary layer blockage, B_{Bcl}. The divergence of the experimental curves from Reneau's curves above AR equal to about 1.5 is due to earlier onset of stall on the low velocity wall for the uniform shear flows. For larger values of AR, as the stall becomes more fully developed, the two sets of curves tend to come together again.

Diffuser Tests With Severely Nonuniform Shear Flows. In evaluating performance for nonuniform shear flows, three sets of diffusers, including the two sets having N/W_1-values of 3 and 6, used in testing the A-profiles, and an additional set having N/W_1 equal to 12, were selected for testing. The long diffusers were added to establish more adequately the effect of mixing on the performance of diffusers with nonuniform shear flows. The geometries of the diffusers include all those marked as x and 0 on the diffuser flow regime chart in Fig. 7.

Three nonuniform shear velocity profiles, designated as B-profiles, were produced with three different velocity-profile generators. The profiles simulate a wake flow, B-1; jet flow, B-2; and step-shear flow, B-3. General characteristics and boundary layer characteristics for these inlet velocity profiles are listed in Table 1. The profiles were designed to have equal gross profile parameters, such as u_{1max}/u_{1min}, B_1, K_{E1}, etc., and thin boundary layers having equivalent boundary layer blockage, B_{B1}. The profiles are compared in Fig. 12. The wake and jet profiles, B-1 and B-2, have flat section in the central part of the profile, indicating that they are still in the "near field" flow region and not yet fully developed. The step-shear profile, B-3, on the other hand, is a fully developed shear layer and can be represented by an error function, such as indicated by Görtler's solution for turbulent shear flows in the "far field" flow region.

Exit velocity profiles were measured for all diffuser tests to facilitate better qualitative interpretation of the performance data. Typical profiles for the $N/W_1 = 6$ diffusers are presented in Figs. 13(*a*), (*b*), and (*c*). The inlet velocity profiles are included as dotted curves for comparison purposes. Fig. 13(*a*) pre-

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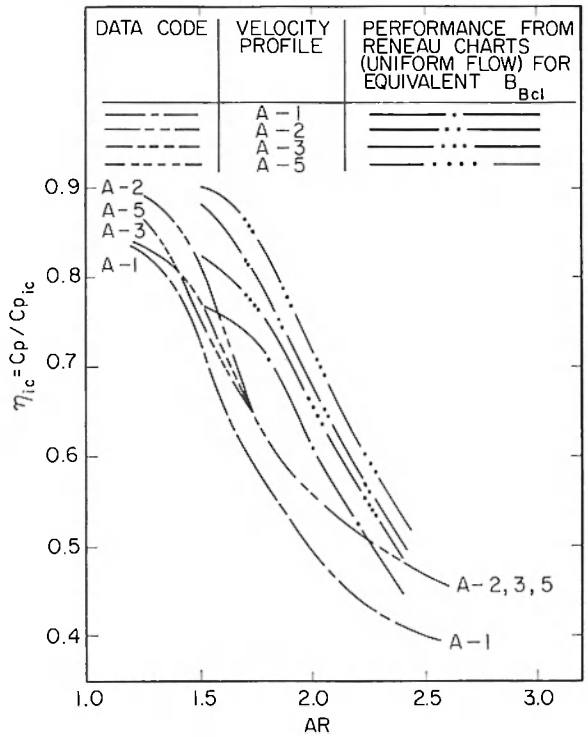
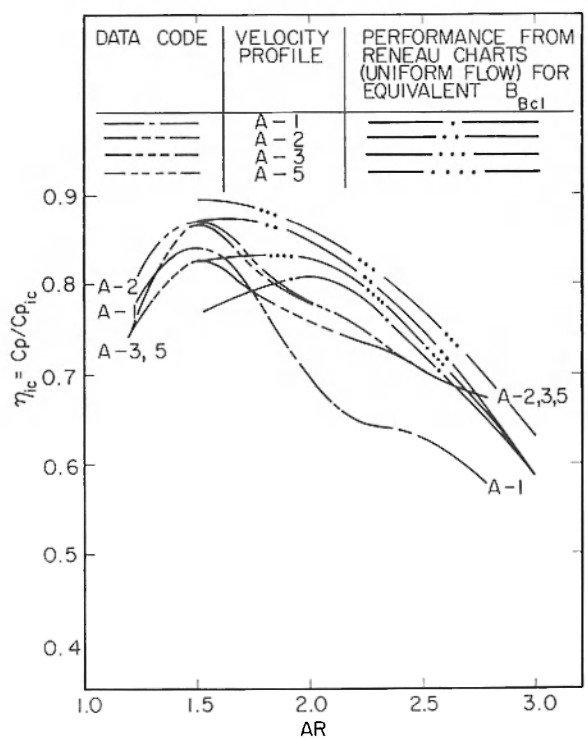


(a) $N/W_1 = 3$



(b) $N/W_1 = 6$

Fig. 11 Comparison of measured effectiveness, η_{ic} of two-dimensional diffusers with simple, uniform shear flows and performance predicted from charts of Reneau, et al. [3]. Note: Curves for A-profiles represent best fit through data taken for AR-values marked on Fig. 7.

sents the wake-flow, B-1, velocity profiles. The significant observations are that the flow started and remained symmetrical and backflow occurred in the central part of the diffuser and not at the walls. The jet-flow, B-2, velocity profiles are presented in

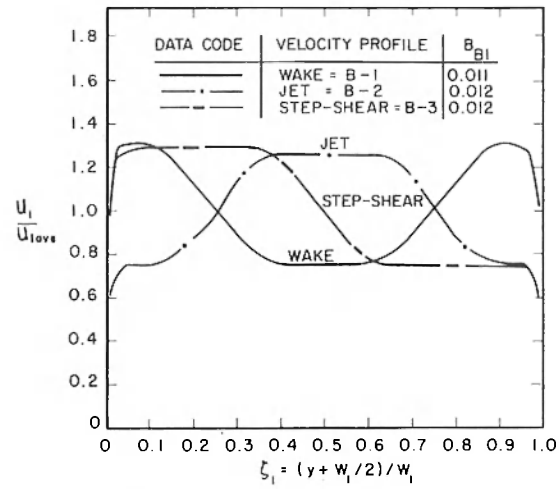


Fig. 12 Measured inlet velocity profiles of severely nonuniform shear flows: B-1, B-2 and B-3. Note: Curves represent best fit through data measured every 0.1 in. in the core and every 0.01 in. in the boundary layer.

Fig. 13(*b*). For the jet flows, the flow started symmetrical but became asymmetric; stall occurred on one wall. The stalled flows were in fact bistable, and the jet could be switched from one wall to the other by partially blocking the flow in the exit plane. Fig. 13(*c*) presents the step-shear flow, B-3, velocity profiles. Here, as expected, wall stall occurred on the low velocity side. The stalled flow was stable and, unlike the jet flow, could not be switched from one wall to the other.

The overall diffuser performance with nonuniform shear flows is tabulated in Table 3. In Figs. 14(*a*), (*b*), and (*c*) the measured Cp-values are individually compared to performance predicted by the ideal diffusion model, Cp_i. Performance curves predicted from the Reneau charts for uniform flow and equivalent inlet boundary layer blockage, B_{Bl}, as well as Cp_{1-D}-curves and Cp_m-curves, based upon models defined in the nomenclature, are included for comparison. For the wake flow, Fig. 14(*a*), it is noteworthy that most of the data lie above the Cp_r-curve and a few points even lie above the corresponding performance curve obtained from the Reneau charts. There is nothing fundamental

Table 3 Performance of diffusers with nonuniform shear flows

Inlet Velocity Profile	Area Ratio AR	Measured Pressure Recovery Cp $N/W_1 = 3$	$N/W_1 = 6$	$N/W_1 = 12$
B-1	1.2	0.245	0.226	---
B-1	1.5	0.453	0.450	0.449
B-1	1.8	0.507	0.535	0.589
B-1	2.1	0.523	0.570	0.643
B-1	2.4	0.499	0.570	---
B-1	2.7	---	0.564	0.680
B-1	3.3	---	---	0.650
B-2	1.2	0.203	0.204	---
B-2	1.5	0.323	0.359	0.435
B-2	1.8	0.302	0.392	0.546
B-2	2.1	0.290	0.382	0.568
B-2	2.4	0.265	0.354	---
B-2	2.7	---	0.335	0.480
B-2	3.3	---	---	0.408
B-3	1.2	0.192	0.171	---
B-3	1.5	0.306	0.314	0.339
B-3	1.8	0.277	0.342	0.379
B-3	2.1	0.257	0.343	0.382
B-3	2.4	0.231	0.336	---
B-3	2.7	---	0.311	0.378
B-3	3.3	---	---	0.361
B-4	1.5	0.451	---	0.475
B-4	2.1	0.494	---	---
B-4	2.7	---	---	0.771
B-4	3.3	---	---	0.779
B-4	2.4	0.477	---	---

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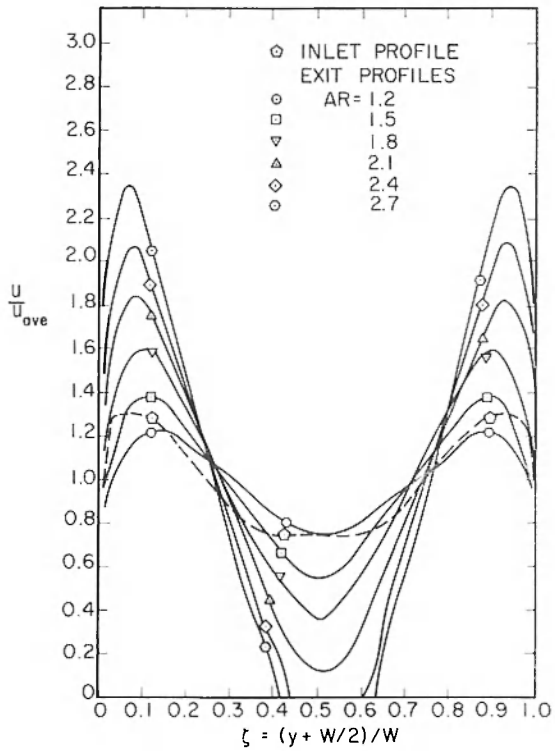

Fig. 13(a) Velocity profile B-1, $N/W_1 = 6$

that prevents such behavior; indeed, it merely indicates that some of the pressure recovery is due to mixing which causes profile flattening. The maximum Cp that can ever be attained is given by the Cp_m-curve, which results from a flow model based on the assumptions of reversible mixing of a nonuniform profile at the inlet to a uniform profile at the exit. As expected, all the data lie well below this theoretical limit. For the jet flow, Fig. 14(*b*), the performance data are generally lower than for the wake flow, indicating that less pressure recovery was obtained by mixing. However, even here for the long diffusers Cp exceeds Cp_i for some AR-values. For the step-shear flow, Fig. 14(*c*), the performance data lie still lower, indicating that even less mixing occurred than for the jet flow. The general decrease in mixing effects represented in decreased Cp-values, can be explained with reference to the core shear layers, which can be regarded as mixing layers. For the wake flow and the jet flow there are two shear layers side by side. However, for the wake flow they are contained within the core region of the flow, while for the jet flow they grow into the wall boundary layer region. Consequently, in the jet-flow case some of the favorable effects of mixing, resulting in profile flattening, are nullified by the unfavorable distortion effect o the wall boundary layer. For the step-shear flow there is only one shear layer, resulting in less mixing than either of the other two flows, and hence less profile flattening and thus lower pressure recovery.

An effectiveness-type of performance prediction method as formulated for the A-profiles, is inadequate for the nonuniform shear flows (B-profiles) since some of the Cp-values are larger than

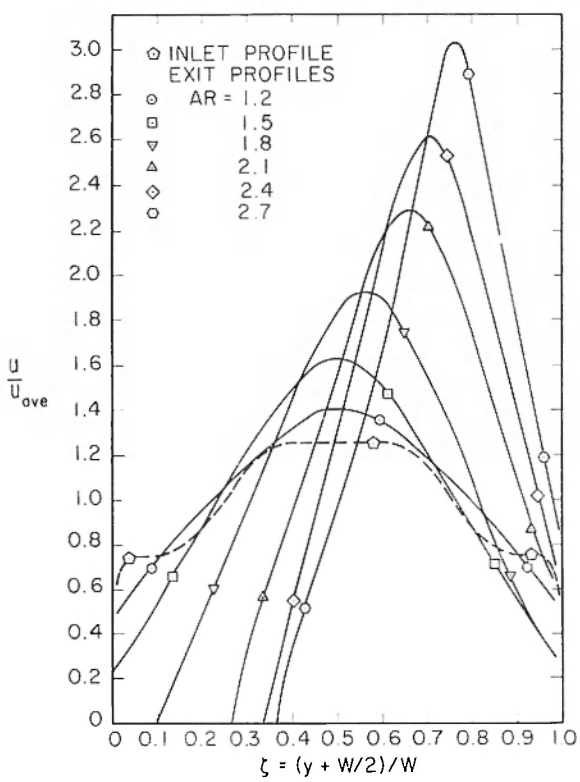

Fig. 13(*b*) Velocity profile B-2, $N/W_1 = 6$

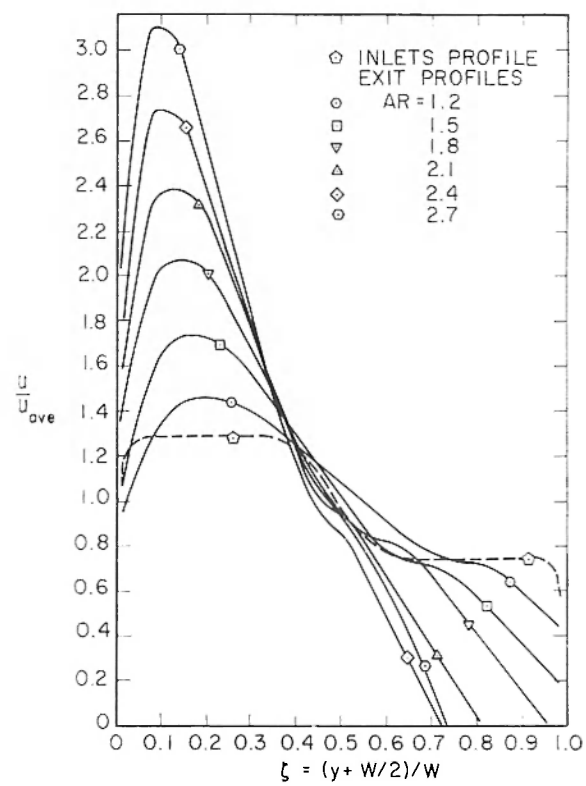

Fig. 13(c) Velocity profile B-3, $N/W_1 = 6$

Fig. 13 Comparison of velocity profiles of severely nonuniform shear flows at inlet and exit sections of two-dimensional diffusers. Note: Exit profile curves represent best fit through data measured every 0.1 in.

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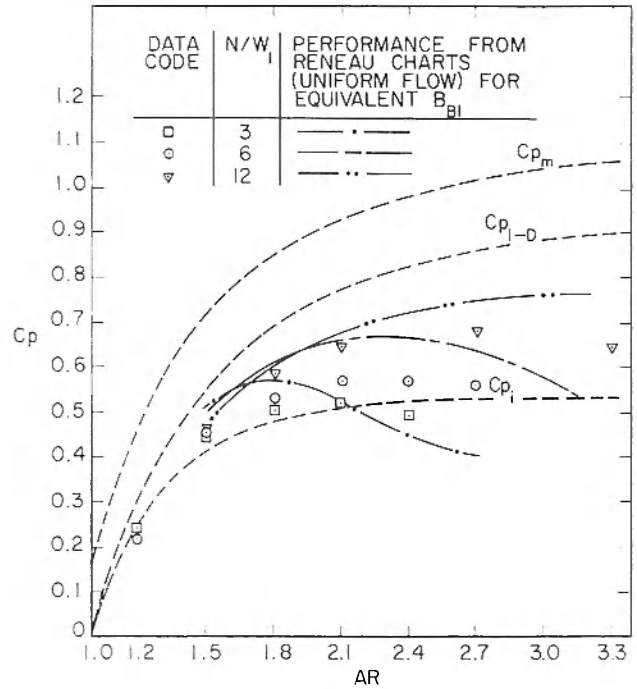

Fig. 14(a) Wake velocity profile B-1, $B_{BI} = 0.011$

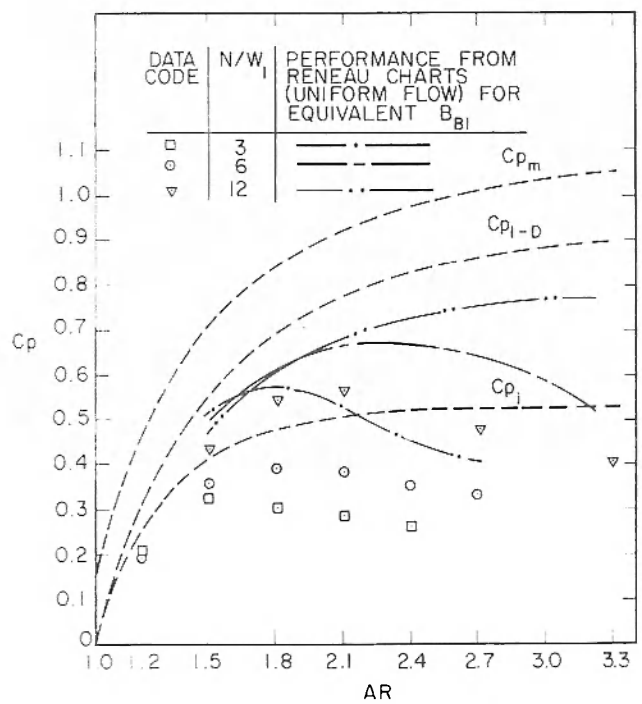

Fig. 14(b) Jet velocity profile B-2, $B_{BI} = 0.012$

Fig. 14 Comparison of measured pressure recovery of two-dimensional diffusers with severely nonuniform shear flows and performance predicted by simple flow models

the Cp_i-values, resulting in effectiveness values, η_G, larger than unity.

Diffuser Tests With Transverse Wake Flow. Because the B-1 wake flow exhibited such good performance in comparison to the jet, B-2, and step-shear, B-3, flows, it was decided to test several diffusers with a different type of wake flow. The wake for the B-1 profile was oriented in the longitudinal (z) direction. A wake flow, designated as B-4, having four equidistantly-spaced transverse (y-direction) wakes, each of the same shape as the B-1 wake profile, was generated.

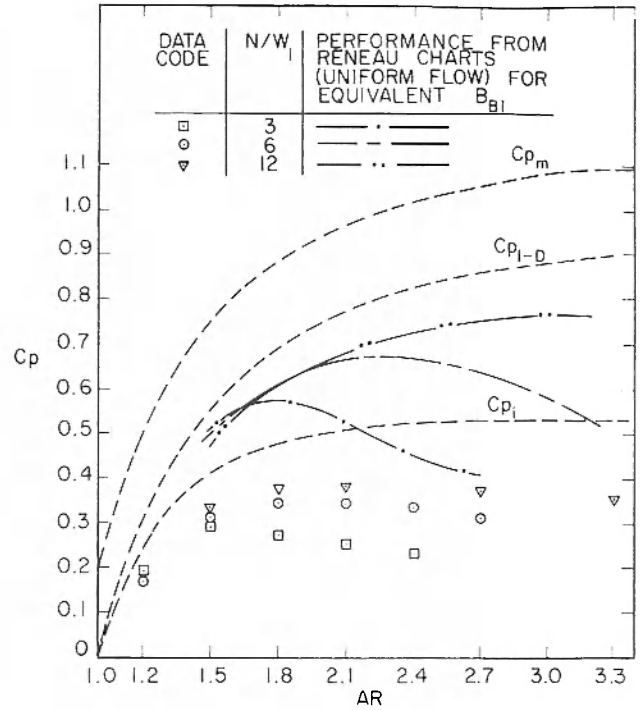

Fig. 14(c) Step-shear velocity profile B-3, $B_{BI} = 0.012$

Tests were conducted on diffusers having N/W_1 equal to 3 and 12. The measured overall pressure recovery, Cp, are listed in Table 3, and in Fig. 15 they are compared to values obtained with B-1 flow. For the $N/W_1 = 3$ diffusers the Cp-values for both B-1 and B-4 flow are about equal. However, for the $N/W_1 = 12$ diffusers there is a significant increase in the Cp-values at high AR for the B-4 flow in comparison to the B-1 flow. This increase must be due to more mixing in the B-4 flow somehow brought

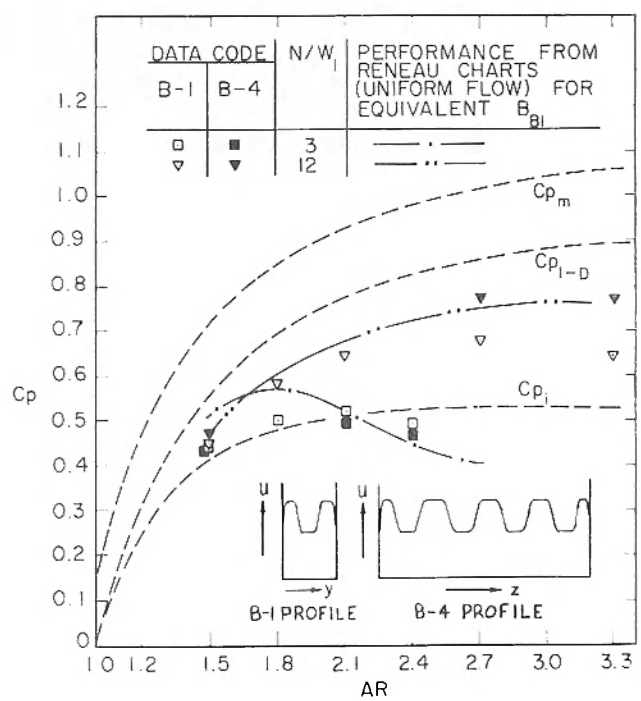

Fig. 15 Comparison of measured pressure recovery of two-dimensional diffusers with transverse wake flow, B-4, and longitudinal wake flow, B-1 (performance predicted by simple flow models applies to B-1 flow)

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about by the interaction of the wakes with the diverging wall boundary layers.

Optimum Geometry and Performance. The diffuser geometries at peak pressure recovery[7] have been correlated to formulate optimum geometry lines for the uniform shear flows, A-profiles; wake flow, B-1; jet flow, B-2; and step-shear flow, B-3. The lines are plotted on an AR-N/W_1 plane in Fig. 16. The numbers adjacent to the data points give the peak pressure-recovery, Cp_{max}, values. The data for the uniform shear flows and jet and step-shear flows have been fitted by a single, straight-line correlation; the line is lower than that found by Reneau, et al. [3] for irrotational core flow. This shift is due to the earlier onset of wall stall, shifting AR for Cp_{max} to a lower value. The wake flow behaves differently than the other nonuniform shear flows. The data-correlating line actually crosses the line of Reneau, et al. at N/W_1 equal to six. This behavior can be attributed to the fact that the wake flow developed *central* stall as opposed to wall stall developed by the other flows.

In Fig. 17 the core distortion effect on peak pressure recovery has been correlated on the basis of an equation derived by Sovran and Klomp [6].

$$Cp = \frac{1}{E_1^2} - \frac{1}{E_2^2 AR^2} \tag{15}$$

The exit effective area fraction, E_2, is plotted as a function of $AR(100B_1)^{1/4}$. The correlating line for present data was fitted by the method of least squares. The line predicts the actual E_2 data within six percent. The correlation found by Sovran and Klomp, from data for irrotational inlet core flow with boundary layer blockage up to 0.05, is included for comparison.

The peak pressure recovery for optimum geometry diffusers having nonuniform flows can be predicted by the following steps: 1, use Fig. 16 to find optimum AR, 2, use Fig. 17 to find the corresponding value of E_2, and 3, use equation (15) to find Cp from AR and E_2. It was found that values of peak recovery predicted by this method were in error up to 31 percent in comparison to the present experimental data for equivalent flow conditions. These large errors are obtained in spite of the good correlation of the data according to the $E_2 - AR(100B_1)^{1/4}$ plot, Fig. 17.

In order to explain the large error cited above, consider the following uncertainty analysis. If E_2 is the only variable considered, the uncertainty, ΔCp, in Cp can be expressed in terms of the uncertainty, ΔE_2, in E_2 by

$$\frac{\Delta Cp}{Cp} = \frac{\partial \ln Cp}{\partial \ln E_2} \frac{\Delta E_2}{E_2}$$

The value of the error multiplication factor $\dfrac{\partial \ln Cp}{\partial \ln E_2} = K$ is obtained by operating on equation (15) to obtain

$$K = \frac{2E_1^2}{AR^2E_2^2 - E_1^2}$$

For large B_1 (maximum B_1 for the nonuniform flows presented in this report is 0.233), the factor K can have a value of order 10. On the other hand, for small B_1 (maximum B_1 in the Sovran and Klomp data correlation is 0.05) the order of the same factor is one. It is because of this error-magnification factor, K, that the Sovran and Klomp correlation works well for flows with small

[7] Peak pressure recovery is defined as the maximum pressure recovery at constant diffuser length, N/W_1.

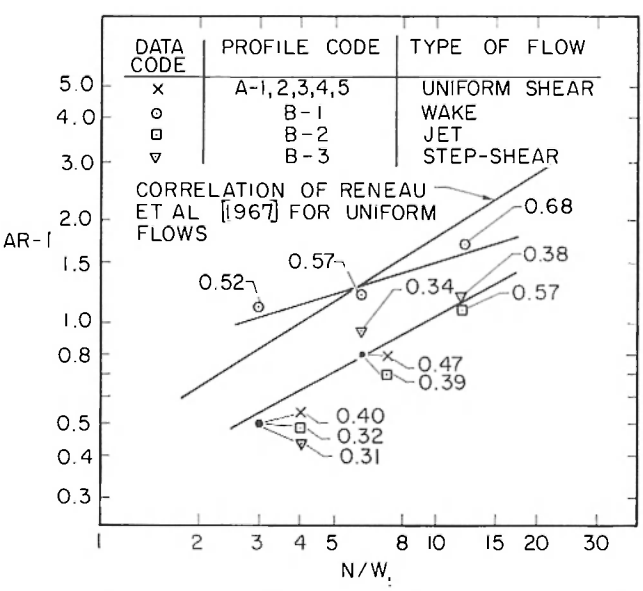

Fig. 16 Geometry at peak pressure recovery at constant N/W_1, (AR-1) as a function of N/W_1. Numbers attached to points indicate observed values of peak pressure recovery.

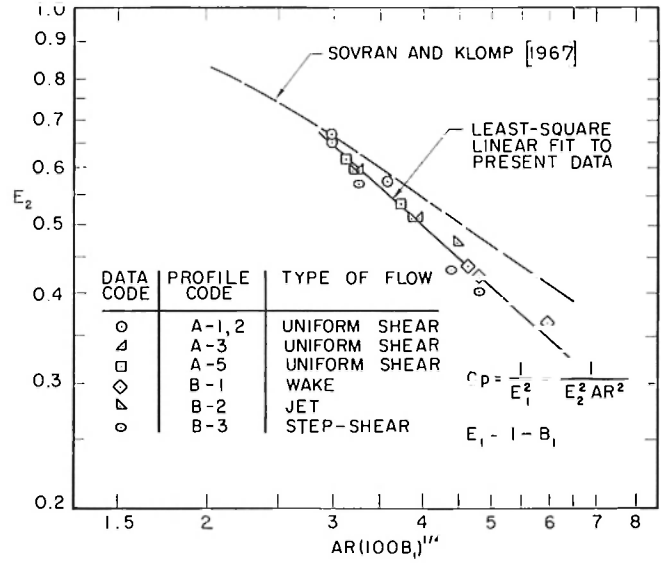

Fig. 17 Correlation for peak pressure recovery at constant N/W_1, E_2 as a function of $AR(100B_1)^{1/4}$

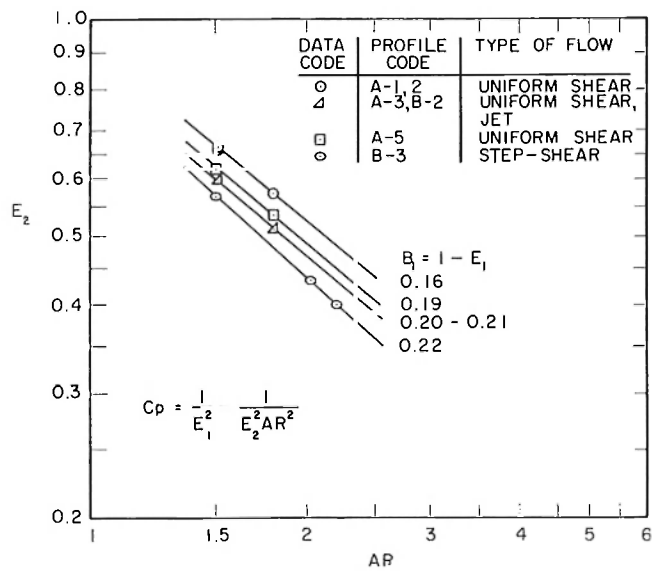

Fig. 18 Correlation for peak pressure recovery at constant N/W_1, E_2 as a function of AR

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B_1-values, but a similar correlation on the *same plane* is inadequate for flows with large B_1-values. Actually, since the correlation of E_0 was based on the premise that viscous forces and mixing in the core region are negligible, all those diffusers that exhibit significant stagnation-pressure losses (above 20 percent of $\rho u_{1avg}^2/2$) on the streamline of maximum velocity should have been eliminated from the correlation. There are four long ($N/W_1 = 12$) diffusers, namely, three with wake flow and one with jet flow, that have significant viscous core losses. However, fitting a line to points of peak pressure recovery (at constant N/W_1) omitting the data for these four diffusers with high core stagnation-pressure loss, did not result in a significantly better correlation.

In Fig. 18, E_2 at peak recovery has been plotted as a function of AR with B_1 as a parameter. The data for the diffusers with high losses of stagnation pressure along the high velocity streamline are not included. The separate line-correlations predict E_2 for the data to within less than one percent, so that the maximum error in predicting Cp-values should be within five percent. A correlation of the type of Fig. 18 is suggested for handling data for large B_1-values from other sources since it shows the greatest promise of giving a reasonable prediction of Cp at peak recovery geometries in terms of a single parameter B_1.

Summary and Conclusions

1 Diffusers with nonuniform inlet velocity profiles having a low velocity core-flow region near one or both walls, such as uniform shear flows (profiles A-1, 2, 3, and 5) and nonuniform shear flows of the jet (profile B-2) and step-shear (profile B-3) type, exhibit similar decreased performance when compared to diffusers having uniform inlet velocity profiles. In particular, the following effects are observed:

(*a*) The onset of stall occurs at a lower value of AR for fixed N/W_1.

(*b*) Once stall has occurred, its features depend markedly on the inlet velocity profile. Diffusers having inlet velocity profiles with high velocity fluid on one side and low velocity fluid on the other such as the uniform shear flows (A-1, 2, 3, and 5) and the step-shear flow (B-3), develop a stable two-dimensional stall on the low velocity side. On the other hand, diffusers having symmetrical inlet velocity profiles with low velocity fluid on both walls, such as the jet flow (B-2), develop a bi-stable two-dimensional stall, which can be switched from one diverging wall to the other by a large disturbance.

2 The performance of diffusers with longitudinal wake flow (B-1) is characterized by several effects:

(*a*) A central stall develops in the core region of the flow (as opposed to wall stall) for diffusers with sufficiently large area ratios.

(*b*) For several geometries, pressure-recovery performance is increased above that measured for diffusers with irrotational inlet core velocity profiles and equal boundary layer blockage.

3 Diffuser performance is affected by the relative location of the low velocity core-flow region in the inlet velocity profile, as observed from tests with the nonuniform shear flows of the wake (B-1), jet (B-2) and step-shear (B-3) type. In general, for equivalent or nearly equivalent total nonuniformity parameters, such as K_{E1}, B_1, $\bar{u}_{1max}/\bar{u}_{1min}$, etc., and equally thin boundary layers, the diffuser performance for these three flows can be arranged in decreasing order as follows: (*a*) the wake flow, (*b*) the jet flow, and (*c*) the step-shear flow.

4 For simple, uniform shear flows, an ideal rotational diffusion model (see equations (11) and (12)) was developed that proved sufficient to define an effectiveness, η_{ic}, which, for *unstalled* flows is approximately equal to the effectiveness determined for uniform core flow with equivalent inlet boundary layer blockage, B_{Bc1}. This conclusion is based on experiments over a limited range in the profile parameter λ (1.50 to 1.73) and boundary layer blockage B_{Bc1} (0.017 to 0.053).

5 For nonuniform shear flows, an ideal rotational diffusion model (see equations (6) and (8)) was developed, relating AR and Cp_i in an integral equation. This model predicts performance trends better than the inviscid one-dimensional model; however, it is inadequate for a precise prediction method, because of mixing within the core region, which is not accounted for in the theory.

6 Optimum geometry lines are plotted on an $AR - N/W_1$ plane in Fig. 16. Data for the uniform shear flows and the jet and step-shear flows are fitted by a single, straight-line correlation; the line is lower than that found by Reneau, et al. [3] for irrotational core flow. The wake flow behaves differently than the other nonuniform shear flows. The data-correlating-line for the wake flow actually crosses the line of Reneau, et al.

7 Peak pressure-recovery performance are correlated on an $E_0 - AR$ plot with B_1 as a parameter, Fig. 18. Cases for which high stagnation-pressure losses occur along the high velocity streamlines (above 20 percent of $\rho u_{1avg}^2/2$) are not included. The maximum pressure recovery of a diffuser flow not violating this limitation can be predicted to within about five percent by using Figs. 16 and 18 and equation (15).

8 Peak pressure-recovery performance are also correlated on an $E_0 - AR(100B_1)^{1/4}$ plot, Fig. 17. All peak recovery data, including those for flows with high stagnation-pressure losses, are included in this correlation. Consequently, the maximum pressure recovery of diffuser flows can be predicted by using Figs. 16 and 17 and equation (15). However, this process is accurate only for inlet flows with values of total inlet blockage factors, B_1, less than about 0.05; for inlet flows with large B_1-values it is recommended *only* in the absence of specified data for the case in hand.

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